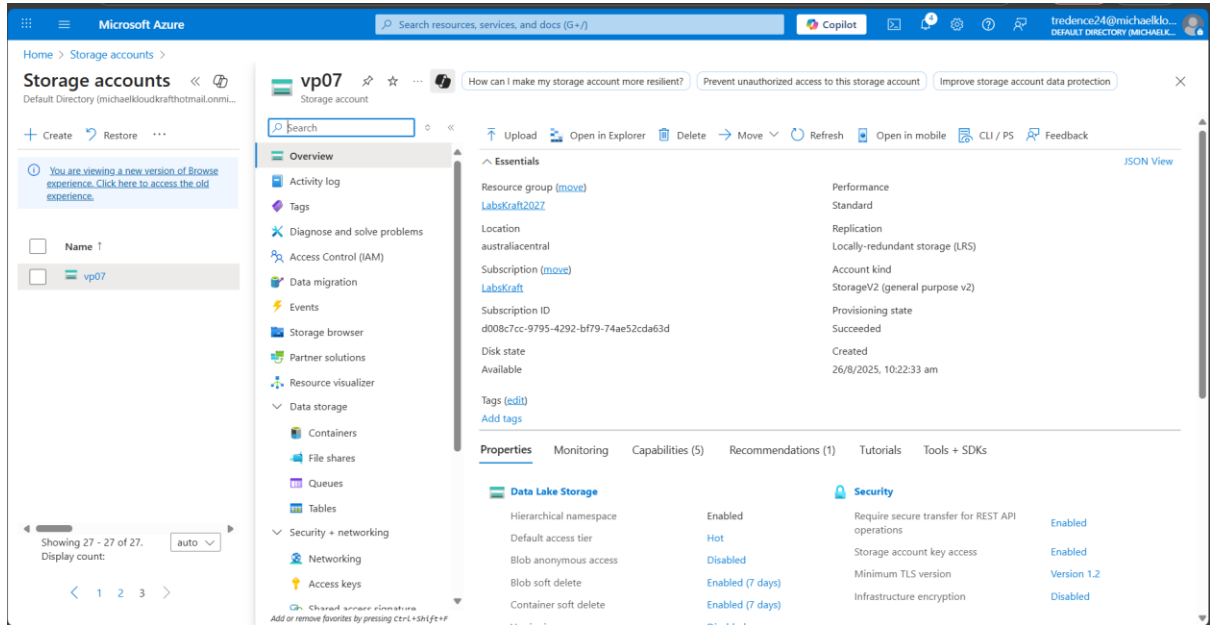


Day 2 Capstone Project Documentation

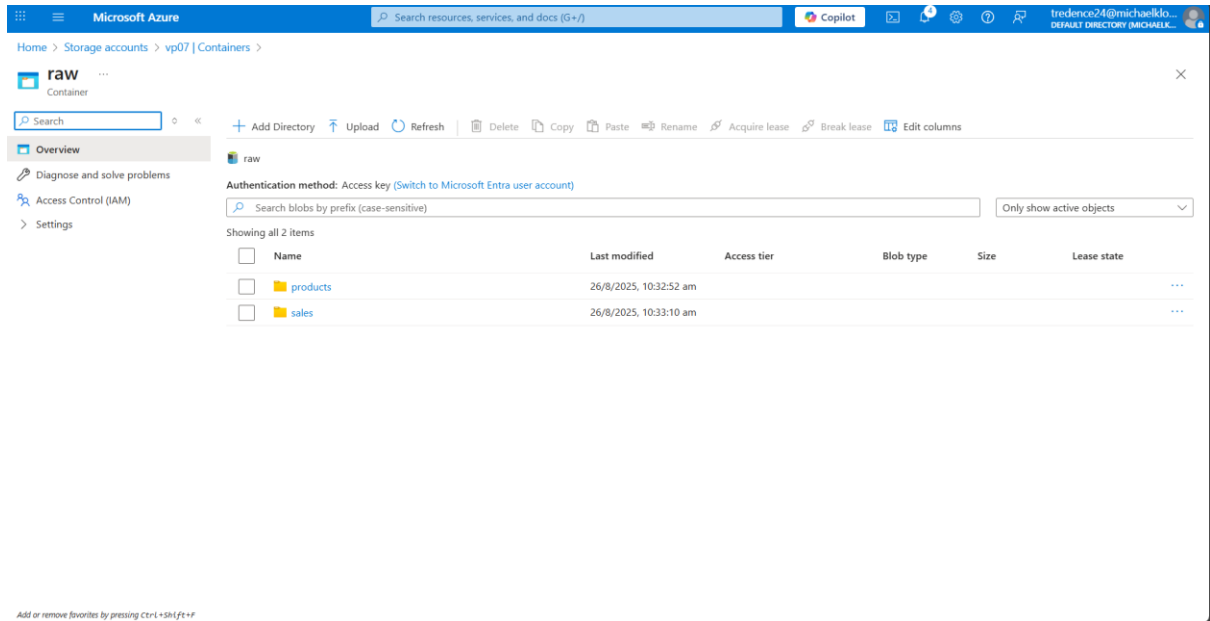
Project 02: Retail Compliance & Analytics Platform on Azure Databricks

1. Create new Storage account



The screenshot displays the Microsoft Azure portal interface for a storage account named 'vp07'. The left sidebar shows the 'Storage accounts' section with a list of accounts, including 'vp07'. The main content area shows the 'Overview' tab for the 'vp07' storage account. The 'Essentials' section lists key properties: Resource group (move), Location (australiacentral), Subscription (move), Subscription ID (d008c7cc-9795-4292-bf79-74ae52cda63d), Disk state (Available), and Tags (edit). The 'Properties' section is expanded, showing 'Data Lake Storage' settings (Hierarchical namespace: Enabled, Default access tier: Hot, Blob anonymous access: Disabled, Blob soft delete: Enabled (7 days), Container soft delete: Enabled (7 days)) and 'Security' settings (Require secure transfer for REST API operations: Enabled, Storage account key access: Enabled, Minimum TLS version: Version 1.2, Infrastructure encryption: Disabled).

2. Create new Container



The screenshot displays the Microsoft Azure portal interface for a container named 'raw'. The left sidebar shows the 'Containers' section with a list of containers, including 'raw'. The main content area shows the 'Overview' tab for the 'raw' container. The 'Authentication method' is set to 'Access key'. The 'Search blobs by prefix (case-sensitive)' field is empty. The 'Only show active objects' dropdown is set to 'Only show active objects'. The 'Showing all 2 items' section displays a table with the following data:

Name	Last modified	Access tier	Blob type	Size	Lease state
products	26/8/2025, 10:32:52 am				...
sales	26/8/2025, 10:33:10 am				...

3. Upload Sales csv file and Product metadata file

The screenshot shows the Microsoft Azure portal interface. The breadcrumb navigation is 'Home > Storage accounts > vp07 | Containers > raw'. The container 'raw' is selected, and the 'Overview' tab is active. The authentication method is 'Access key'. A search bar is present with the text 'Search blobs by prefix (case-sensitive)'. Below the search bar, it says 'Showing all 2 items'. A table lists the items in the container:

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	[-]						...
☐	products.csv	26/8/2025, 10:42:52 am	Hot (Inferred)	Block blob	36.69 KiB	Available	...
☐	products.json	26/8/2025, 10:42:52 am	Hot (Inferred)	Block blob	138.2 KiB	Available	...

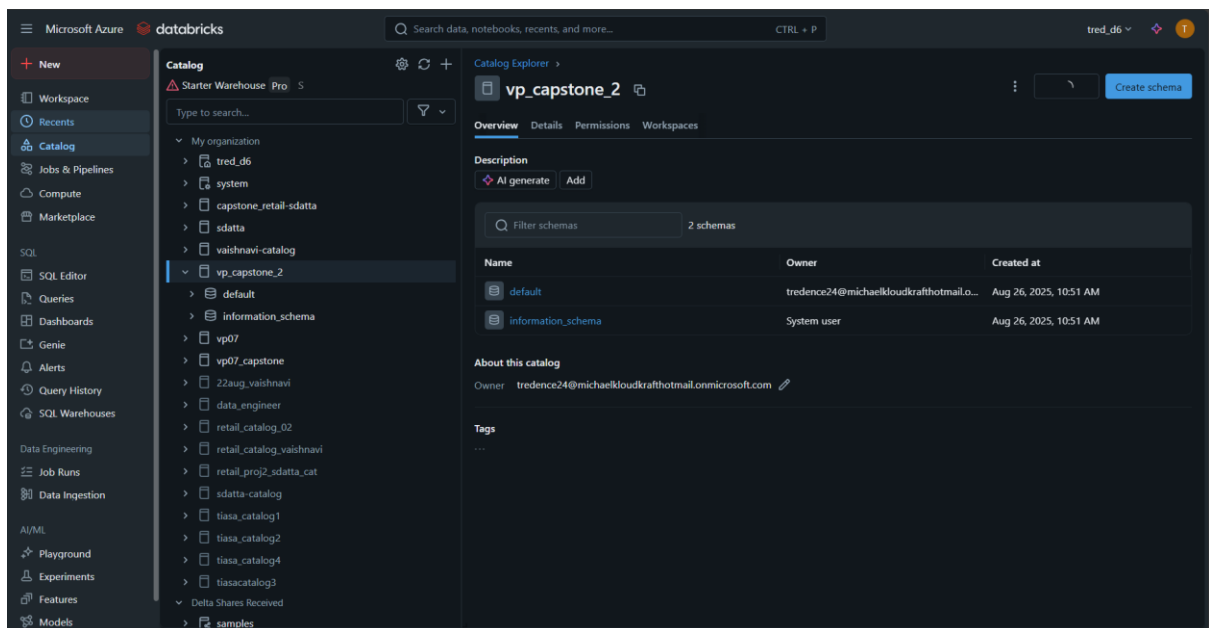
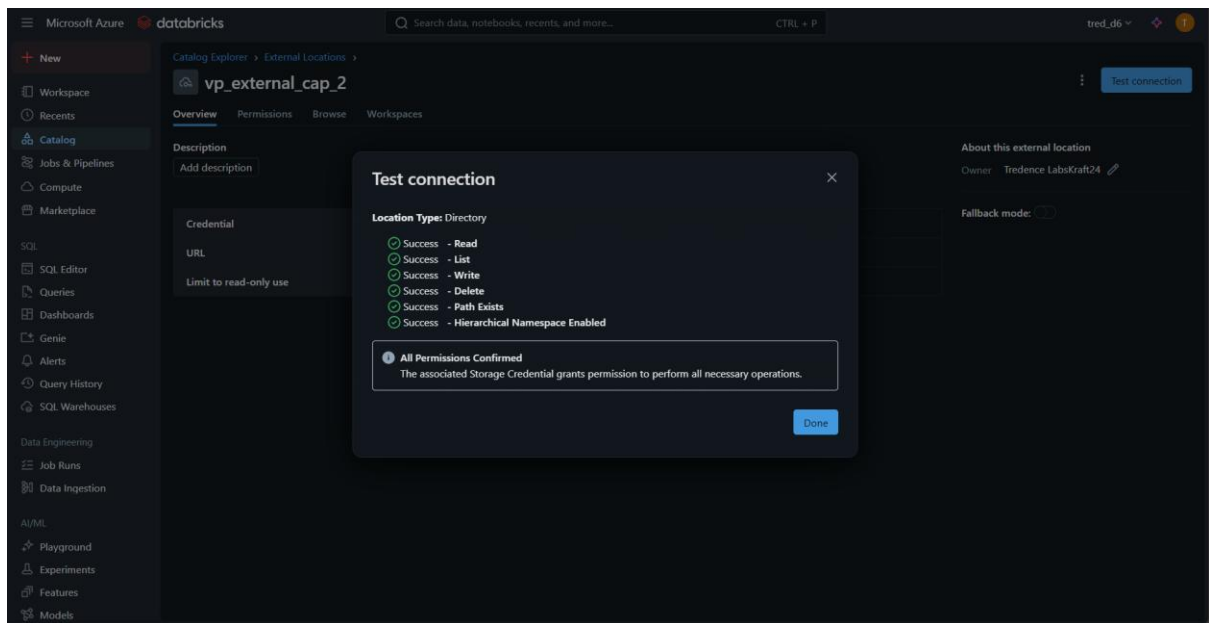
At the bottom of the page, there is a note: 'Add or remove favorites by pressing Ctrl+Shift+F'.

The screenshot shows the Microsoft Azure portal interface. The breadcrumb navigation is 'Home > Storage accounts > vp07 | Containers > raw'. The container 'raw' is selected, and the 'Overview' tab is active. The authentication method is 'Access key'. A search bar is present with the text 'Search blobs by prefix (case-sensitive)'. Below the search bar, it says 'Showing all 5 items'. A table lists the items in the container:

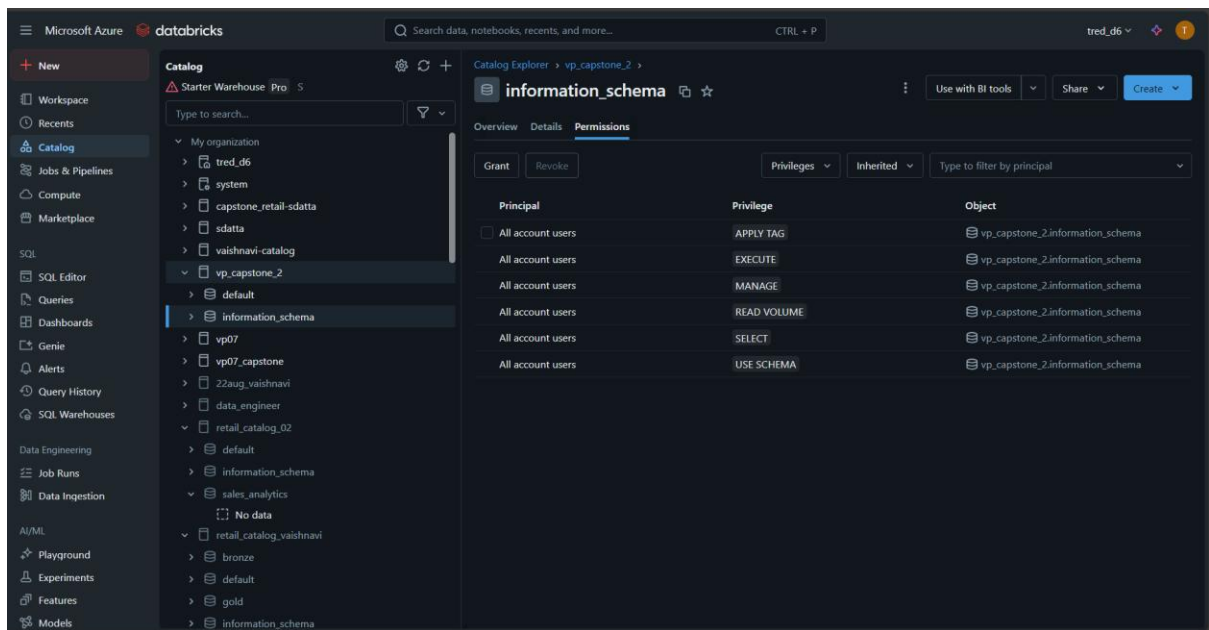
☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	[-]						...
☐	sales_day_1.csv	26/8/2025, 10:43:14 am	Hot (Inferred)	Block blob	50.54 KiB	Available	...
☐	sales_day_2.csv	26/8/2025, 10:43:14 am	Hot (Inferred)	Block blob	50.48 KiB	Available	...
☐	sales_day_3.csv	26/8/2025, 10:43:14 am	Hot (Inferred)	Block blob	50.46 KiB	Available	...
☐	sales_day_4.csv	26/8/2025, 10:43:14 am	Hot (Inferred)	Block blob	50.5 KiB	Available	...
☐	sales_day_5.csv	26/8/2025, 10:43:14 am	Hot (Inferred)	Block blob	50.51 KiB	Available	...

At the bottom of the page, there is a note: 'Add or remove favorites by pressing Ctrl+Shift+F'.

4. Create Azure Connector and add IAM roles, and create a new catalogue in databricks



5. Create new schema



6. Code for DLT pipeline

```
%sql
CREATE SCHEMA IF NOT EXISTS vp_capstone_2.sales_analytics

%sql
USE CATALOG vp_capstone_2;
USE SCHEMA sales_analytics;

%sql
CREATE EXTERNAL TABLE IF NOT EXISTS vp_capstone_2.sales_analytics.raw_sales (
  sale_id INT,
  product_id INT,
  region STRING,
  store_id STRING,
  quantity INT,
  sales_amount DOUBLE,
  sale_date TIMESTAMP
)
USING CSV
OPTIONS (
  header = 'true',
  inferSchema = 'false'
)
LOCATION 'abfss://rawcp2@vp07.dfs.core.windows.net/sales/';
```

```
%sql
CREATE EXTERNAL TABLE IF NOT EXISTS vp_capstone_2.sales_analytics.raw_products (
  product_id INT,
  product_name STRING,
  category STRING,
  price DOUBLE,
  discount_rate DOUBLE
)
USING CSV
OPTIONS (
  header = 'true',
  inferSchema = 'true'
)
LOCATION 'abfss://rawcp2@vp07.dfs.core.windows.net/products/';
```

```
import dlt
from pyspark.sql.functions import col, to_date, sum, countDistinct
from pyspark.sql.types import StructType, StructField, StringType, IntegerType,
DoubleType, DateType

# SALES SCHEMA
sales_schema = StructType([
  StructField("sale_id", StringType(), True),
  StructField("product_id", StringType(), True),
  StructField("region", StringType(), True),
  StructField("store_id", StringType(), True),
  StructField("quantity", IntegerType(), True),
  StructField("sales_amount", DoubleType(), True),
  StructField("sale_date", StringType(), True)
])

# PRODUCTS SCHEMA (common structure for both CSV and JSON files)
products_schema = StructType([
  StructField("product_id", StringType(), True),
  StructField("product_name", StringType(), True),
  StructField("category", StringType(), True),
  StructField("price", DoubleType(), True),
  StructField("discount_rate", DoubleType(), True)
])
```

```

# -----
# BRONZE LAYER
# -----

@dlt.table(name="bronze_sales")
def bronze_sales():
    return (
        spark.readStream.format("cloudFiles")
            .option("cloudFiles.format", "csv")
            .schema(sales_schema)
            .load("abfss://rawcp2@vp07.dfs.core.windows.net/sales/")
    )

# Read Products CSV
@dlt.table(name="bronze_products_csv")
def bronze_products_csv():
    return (
        spark.readStream.format("cloudFiles")
            .option("cloudFiles.format", "csv")
            .schema(products_schema)
            .load("abfss://rawcp2@vp07.dfs.core.windows.net/products/*.csv")
    )

# Read Products JSON
@dlt.table(name="bronze_products_json")
def bronze_products_json():
    return (
        spark.readStream.format("cloudFiles")
            .option("cloudFiles.format", "json")
            .schema(products_schema) # explicitly provide schema (fixes error)
            .load("abfss://rawcp2@vp07.dfs.core.windows.net/products/*.json")
    )

# -----
# SILVER LAYER (Cleaned)
# -----

@dlt.table(name="silver_sales")
@dlt.expect("Valid sales amount", "sales_amount >= 0")
@dlt.expect("Valid quantity", "quantity > 0")
def silver_sales():
    df = dlt.read("bronze_sales")
    return (
        df.dropna()
            .withColumn("sales_amount", col("sales_amount").cast("double"))
            .withColumn("quantity", col("quantity").cast("int"))
    )

```

```

        .withColumn("sale_date", to_date("sale_date", "yyyy-MM-dd"))
    )

# Union both product sources
@dlt.table(name="silver_products")
def silver_products():
    csv_df = dlt.read("bronze_products_csv")
    json_df = dlt.read("bronze_products_json")
    return (
        csv_df.unionByName(json_df)
        .dropna()
        .dropDuplicates(["product_id"])
    )

# -----
# GOLD LAYER
# -----

@dlt.table(name="gold_sales_summary")
def gold_sales_summary():
    sales = dlt.read("silver_sales")
    products = dlt.read("silver_products")

    joined_df = sales.join(products, on="product_id", how="inner")

    result = (
        joined_df
        .withColumn("effective_price", col("price") * (1 - col("discount_rate")))
        .withColumn("discount_amount", col("price") * col("discount_rate") *
col("quantity"))
        .groupBy("sale_date", "region", "product_id", "product_name", "category")
        .agg(
            sum("quantity").alias("total_units_sold"),
            sum("sales_amount").alias("total_revenue"),
            sum("discount_amount").alias("total_discount_given"),
            sum("effective_price").alias("total_effective_price")
        )
    )

    return result

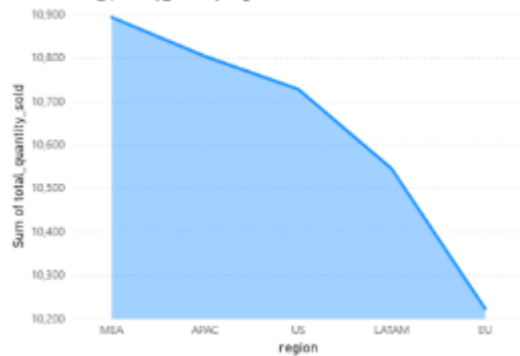
```

7. DLT Pipeline



8. Visualize using Power Bi

Sum of total_quantity_sold by region



Sum of total_quantity_sold by region

